Outline

Optimization

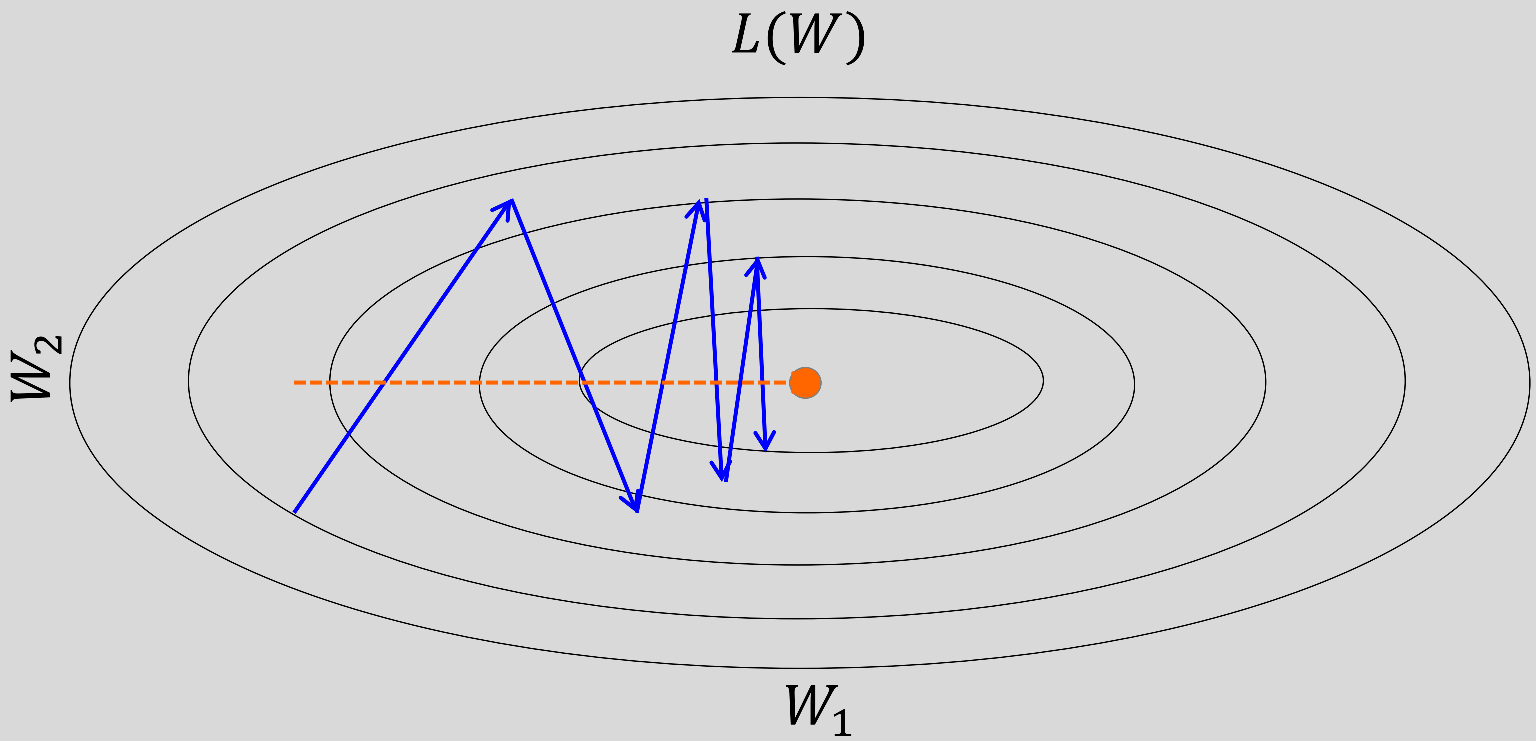
- Challenges in Optimization
- **Momentum**
- Adaptive Learning Rate
- Parameter Initialization
- Batch Normalization

Regularization of NN

- Norm Penalties
- Early Stopping
- Data Augmentation
- Sparse Representation
- Dropout

Momentum

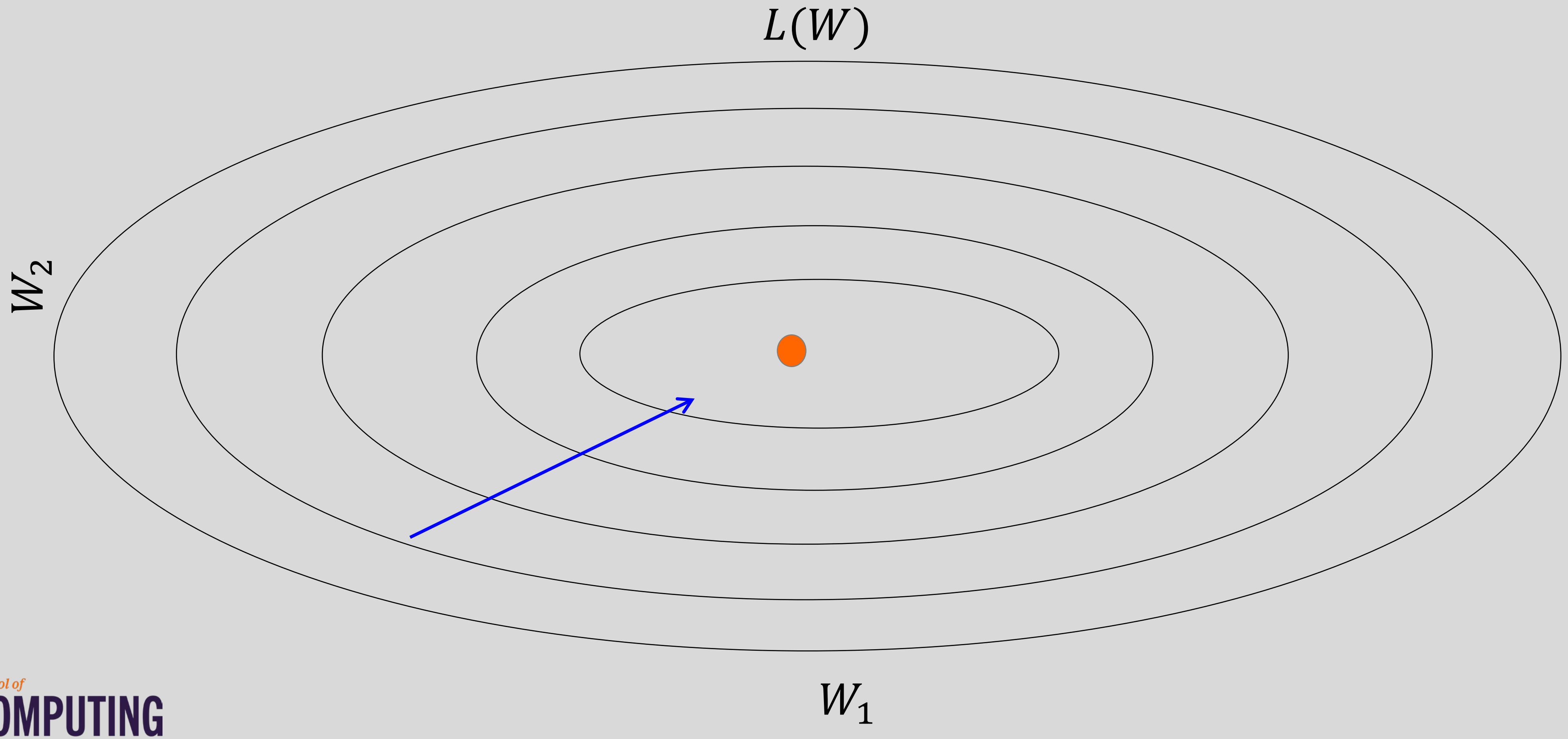
Oscillations because updates do not exploit curvature information



Average gradient presents faster path to optimal:
vertical components cancel out

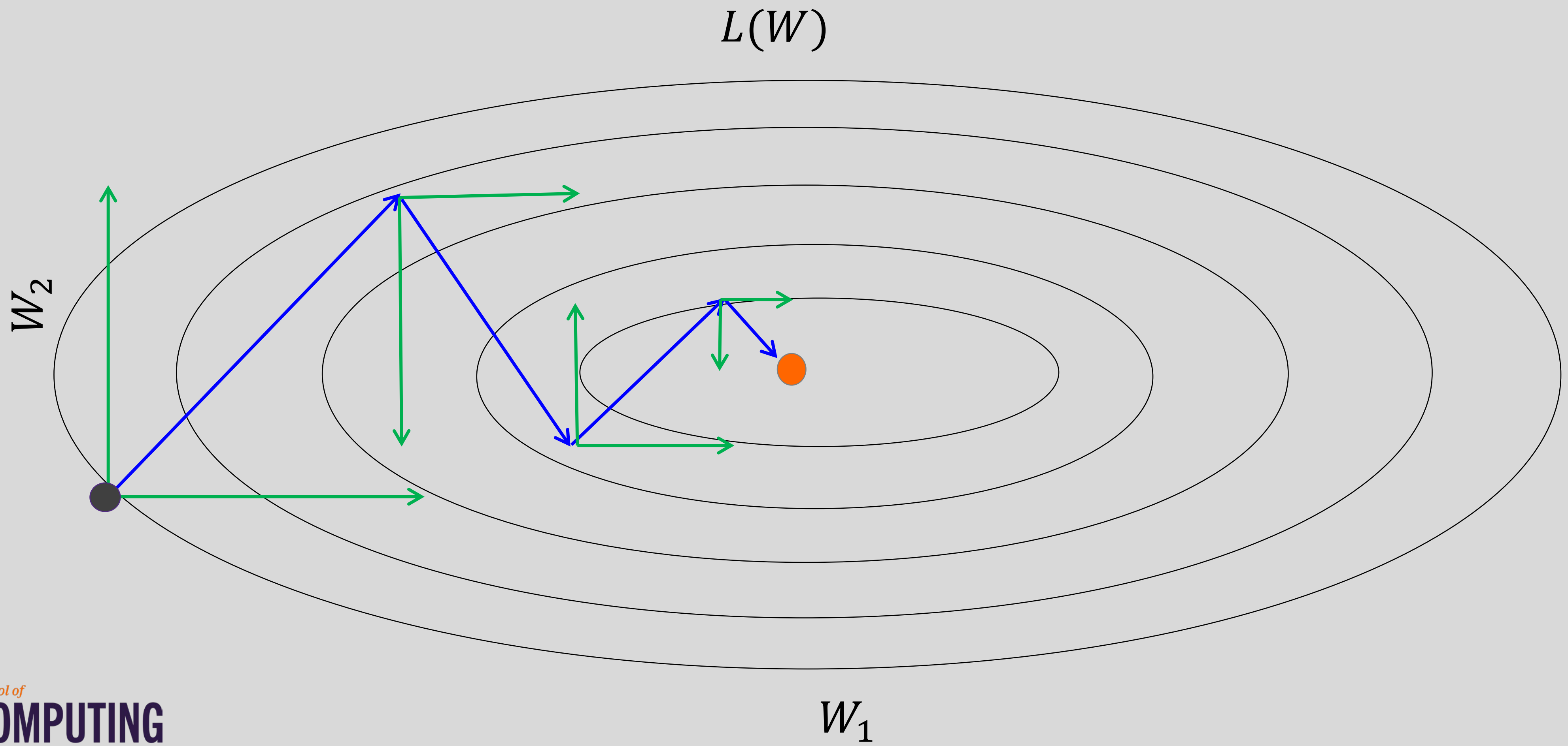
Momentum

Question: Why not this?



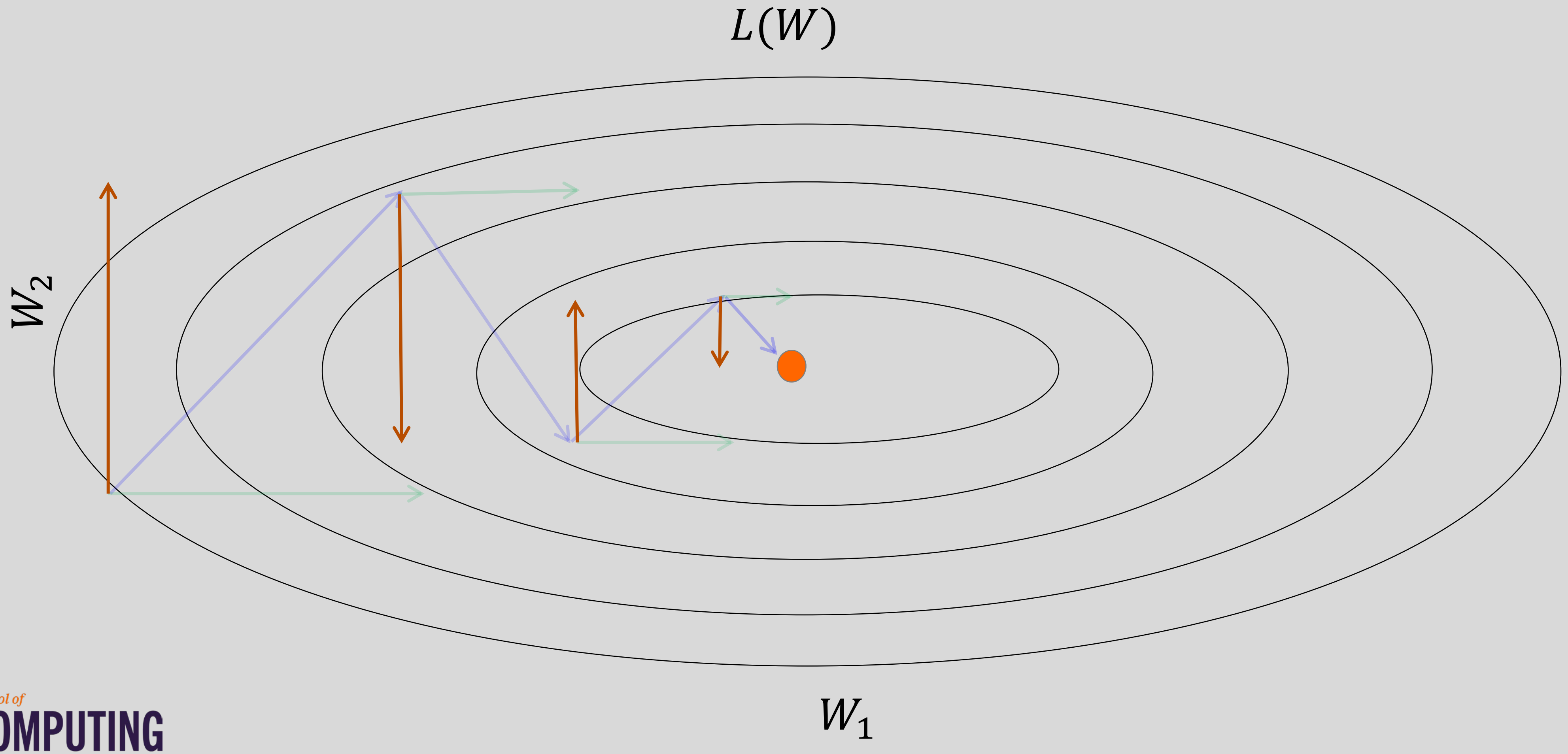
Momentum

Let us figure out an algorithm which will lead us to the minimum faster.



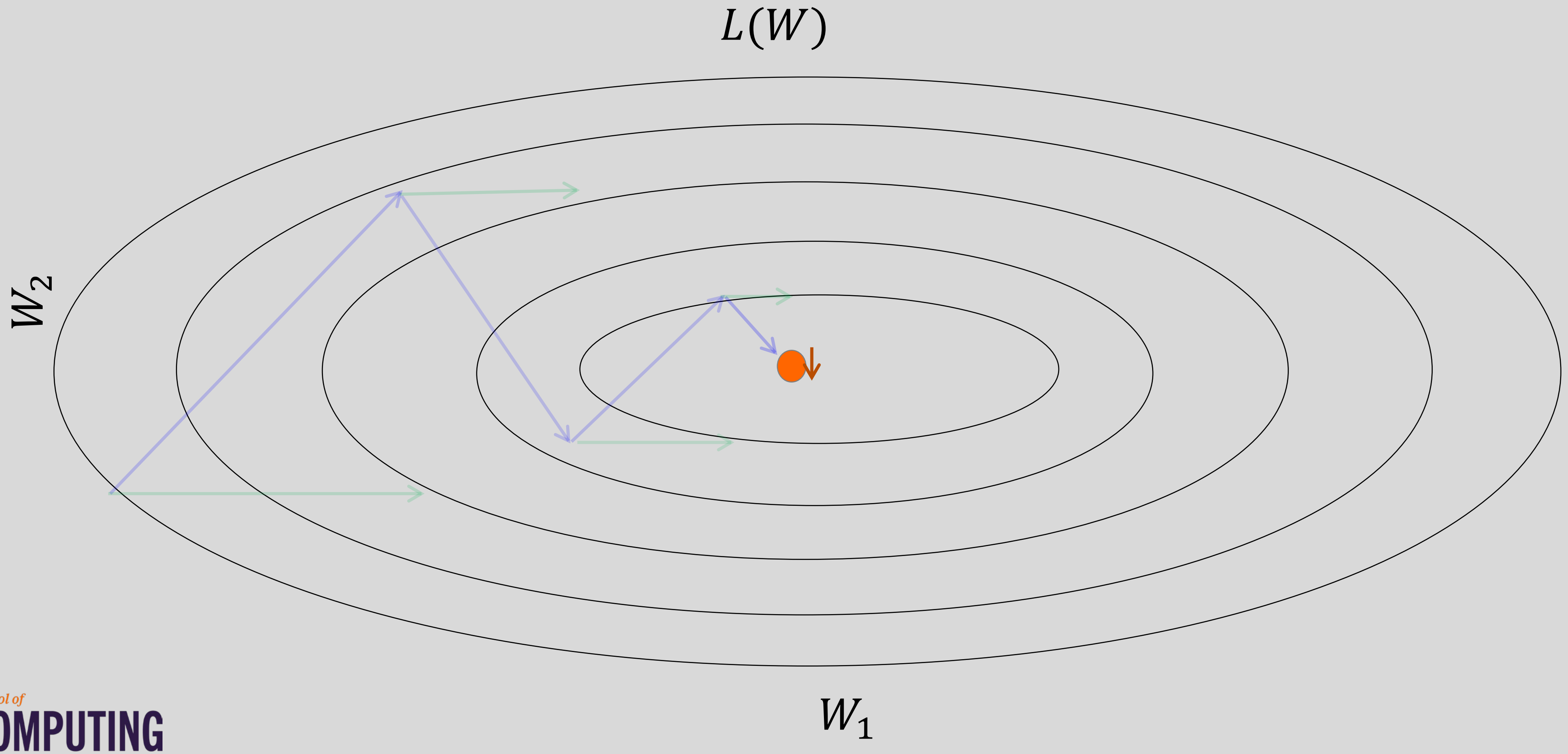
Momentum

Look each component at a time



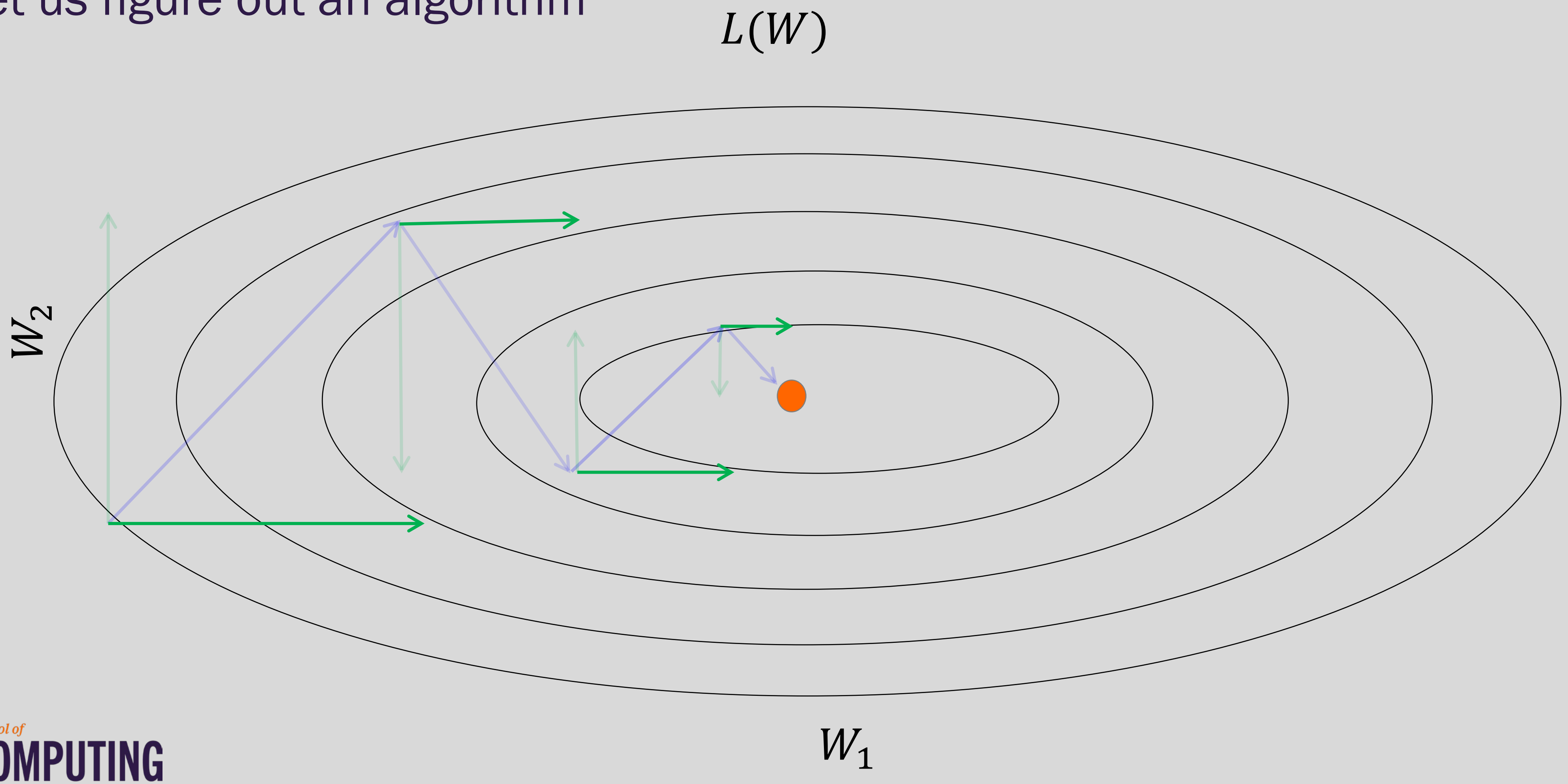
Momentum

Let us figure out an algorithm



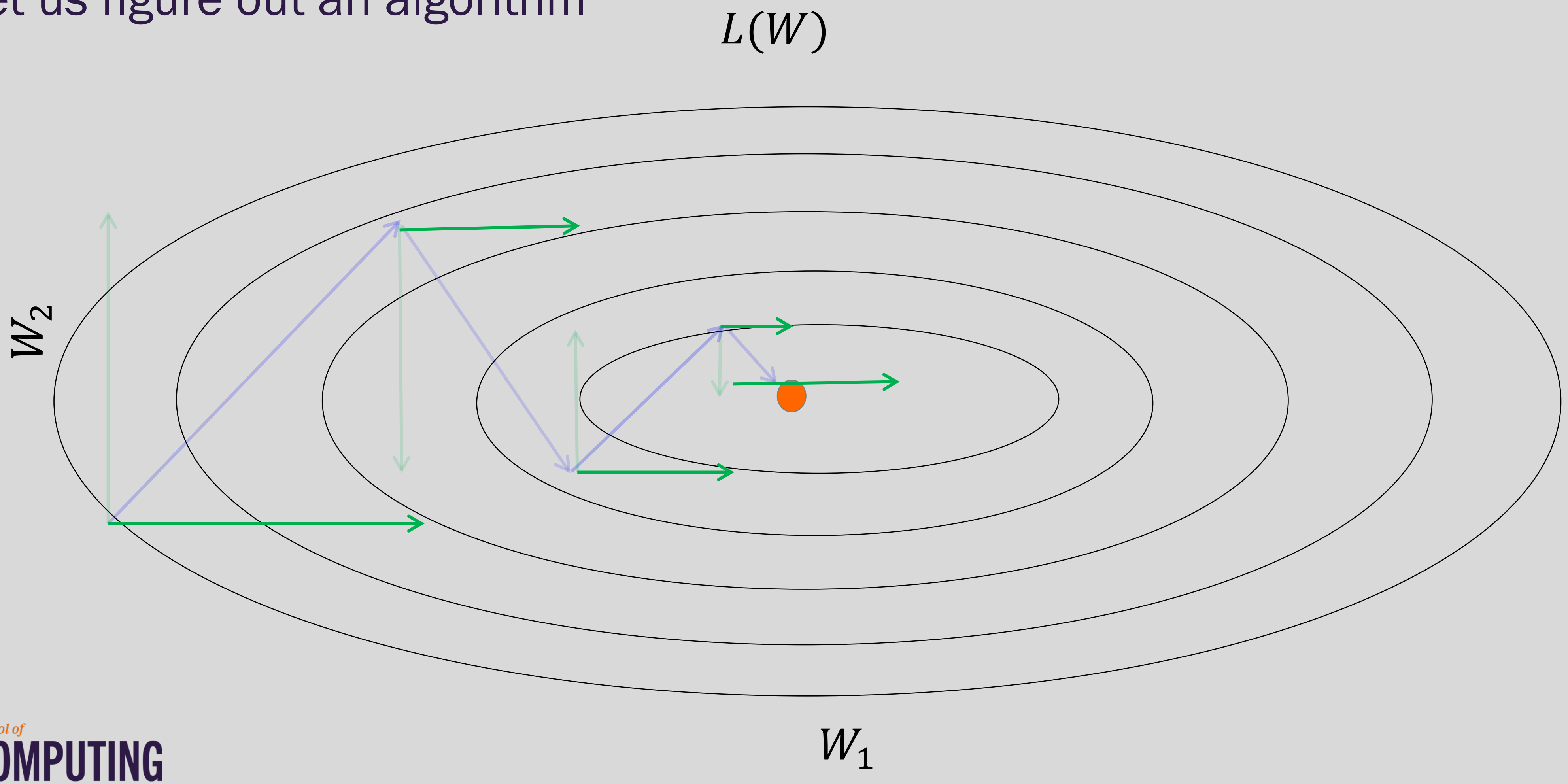
Momentum

Let us figure out an algorithm



Momentum

Let us figure out an algorithm



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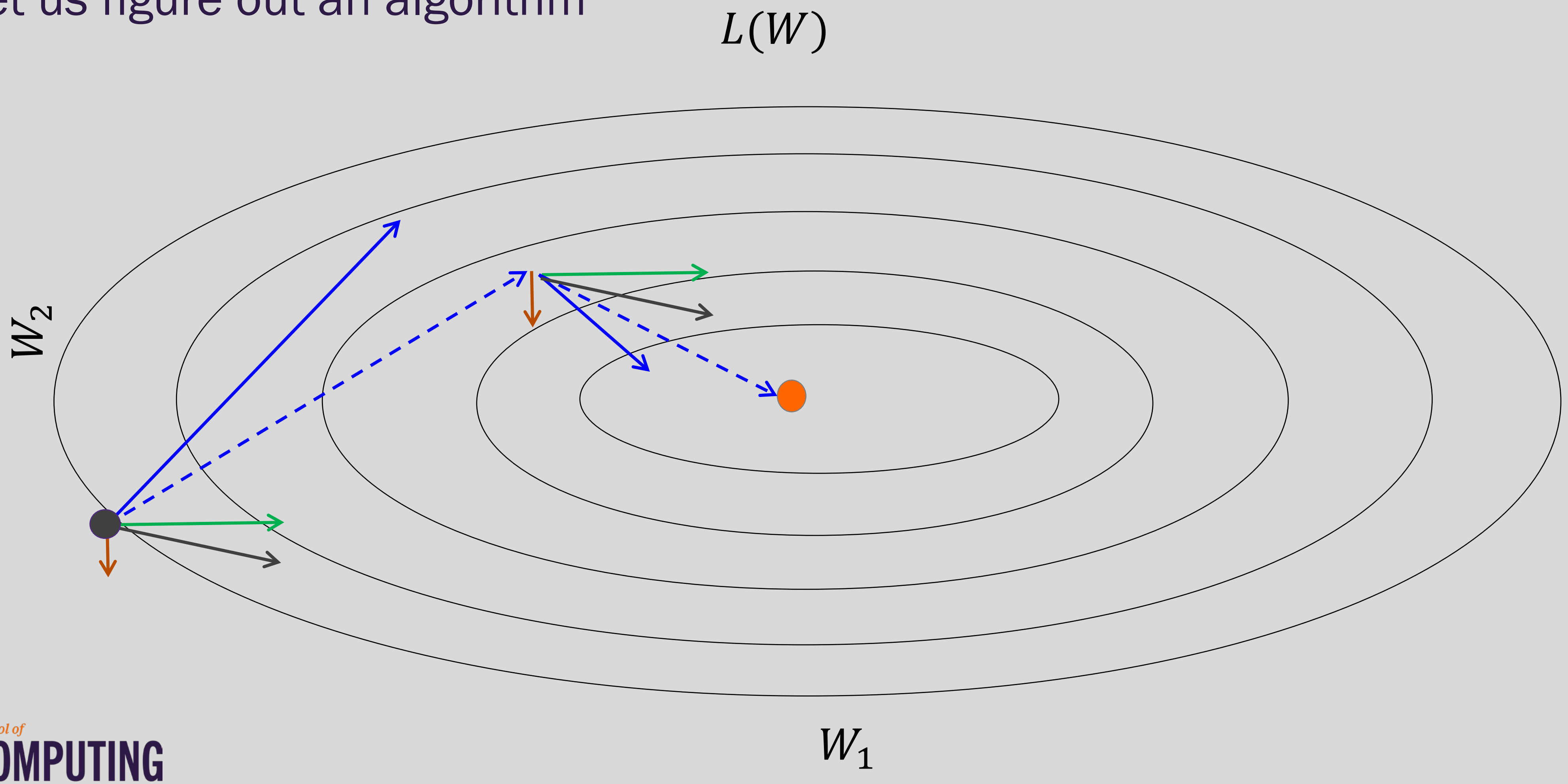
Momentum

Let us figure out an algorithm



Momentum

f is the Neural Network

Old gradient descent:

$$g = \frac{1}{m}\sum_i \nabla_W L(f(x_i; W), y_i) \qquad W^* = W - \lambda g$$

$$g = g + average\ g\ from\ before$$

New gradient descent with momentum:

$$v = \alpha v + (1 - \alpha)\, g \qquad W^* = W - \lambda v$$

$\alpha \in [0,1)$ controls how quickly effect of past gradients decay

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Nesterov Momentum

Apply an interim update: $\tilde{W} = W + v$

Perform a correction based on gradient at the interim point:

$$g = \frac{1}{m}\sum_i \nabla_W L(f(x_i;\tilde{W}), y_i)$$

$$v = \alpha v - \varepsilon g$$

$$W = W + v$$

Momentum based on look-ahead slope